

Lecture-16

Mathematics 1 (15B11MA111)

Module: Double Integrals

CO [C105.3]

Change of Variables in Integrals

Let the variables x, y is defined in the region R of the x - y plane be transformed as $x=x(u,v)$, $y=y(u,v)$. We assume that

- $x(u,v)$, $y(u,v)$ are defined and have continuous partial derivatives in the region R^* of interest in the u - v plane.
- Inverse functions $u=u(x,y)$, $v=v(x,y)$ are defined and are continuous in the region of interest in the x - y plane.

Since the function $f(x,y)$ is continuous in R the function

$f[(x(u,v), y(u,v))]$ is also continuous in R^* .

Topics to be covered

➤ Change of Variables in Integrals:

- ❖ Definition
- ❖ Examples
- ❖ Assignment

➤ Reference for the lecture

R.K Jain and S.R.K. Iyenger, “Advanced Engineering Mathematics” fifth edition, Narosa publishing house, 2016.

Then the double integral transforms as

$$\int \int_R f(x, y) dx dy = \int \int_{R^*} f[(x(u, v), y(u, v))] |J| du dv = \int \int_{R^*} F(u, v) du dv$$

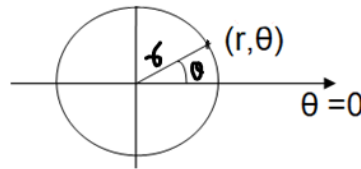
Where $J = \frac{\partial(x, y)}{\partial(u, v)} = \begin{vmatrix} \partial x / \partial u & \partial x / \partial v \\ \partial y / \partial u & \partial y / \partial v \end{vmatrix}$

is the Jacobian of the variables of transformation.

Special Case

Changing Cartesian coordinates to polar coordinates

$$x = r \cos \theta, y = r \sin \theta, r_1 \leq r \leq r_2, \theta_1 \leq \theta \leq \theta_2$$



$$J = \frac{\partial(x, y)}{\partial(r, \theta)} = \begin{vmatrix} \partial x / \partial r & \partial x / \partial \theta \\ \partial y / \partial r & \partial y / \partial \theta \end{vmatrix} = r$$

$$\iint_R f(x, y) dx dy = \iint_{R^*} f(r \cos \theta, r \sin \theta) r dr d\theta = \iint_{R^*} F(r, \theta) r dr d\theta$$

Where R^* is the region corresponding to R in the $r - \theta$ plane.

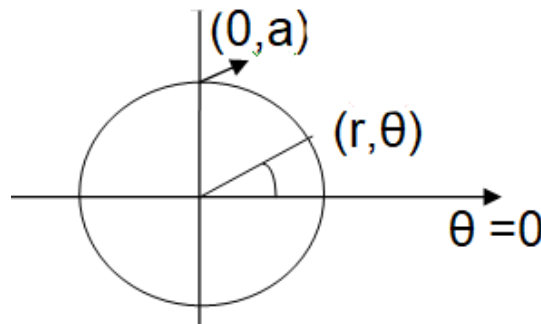
Examples

Example1: Evaluate by changing to polar coordinates

$$\int_{x=0}^a \int_{y=0}^{\sqrt{a^2-x^2}} y^2 \sqrt{x^2+y^2} dy.dx$$

Solution: From upper limit of y we have,

$y = \sqrt{a^2 - x^2} \Rightarrow x^2 + y^2 = a^2$ (which is a circle of radius 'a' and centre at (0,0)). For polar coordinates, $x = a\cos\theta$; $y = a\sin\theta$



$$\begin{aligned}
 I &= \int_{x=0}^a \int_{y=0}^{\sqrt{a^2-x^2}} y^2 \sqrt{x^2+y^2} dy dx \\
 &= \int_{\theta=0}^{\pi/2} \int_{r=0}^a r^2 \sin^2 \theta \cdot r \cdot r dr d\theta
 \end{aligned}$$

$$\begin{aligned}
 I &= \int_{\theta=0}^{\pi/2} \sin^2 \theta \left(\int_{r=0}^a r^4 dr \right) d\theta \\
 &= \frac{a^5}{5} \int_{\theta=0}^{\pi/2} \sin^2 \theta d\theta
 \end{aligned}$$

Using the definition of gamma function

$$\frac{a^5}{5} \int_{\theta=0}^{\pi/2} \sin^2 \theta d\theta = \frac{a^5}{5} \frac{\left(\frac{(2+1)}{2} \right) \left(\frac{(0+1)}{2} \right)}{2 \left(\frac{(2+0+2)}{2} \right)} = \frac{a^5}{5} * 1/2 * \pi/2 = \frac{a^5 \pi}{20}$$

Therefore,

$$I = \int_{x=0}^a \int_{y=0}^{\sqrt{a^2-x^2}} y^2 \sqrt{x^2 + y^2} dy dx = \frac{a^5 \pi}{20}$$

Example 2: Using the transformation $x + y = u; y = uv$ show that

$$\int_0^1 \int_0^{1-x} e^{\frac{y}{x+y}} dy dx = \frac{e-1}{2}$$

Solution:

$$x + y = u; y = uv \text{ therefore } x = u(1 - v)$$

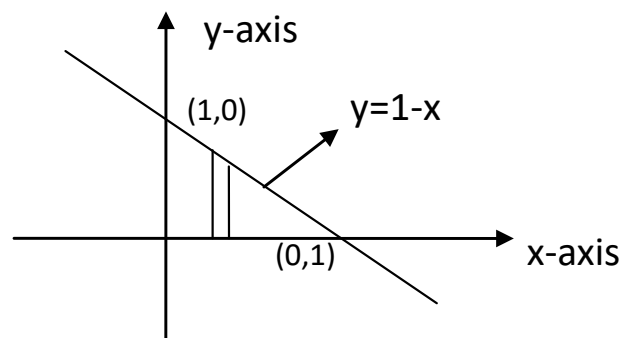
$$\text{Now } y = 0 \Rightarrow uv=0 \text{ (either } u \text{ is zero or } v \text{ is zero)}$$

$$y = 1-x \Rightarrow x+y=1 \Rightarrow u = 1$$

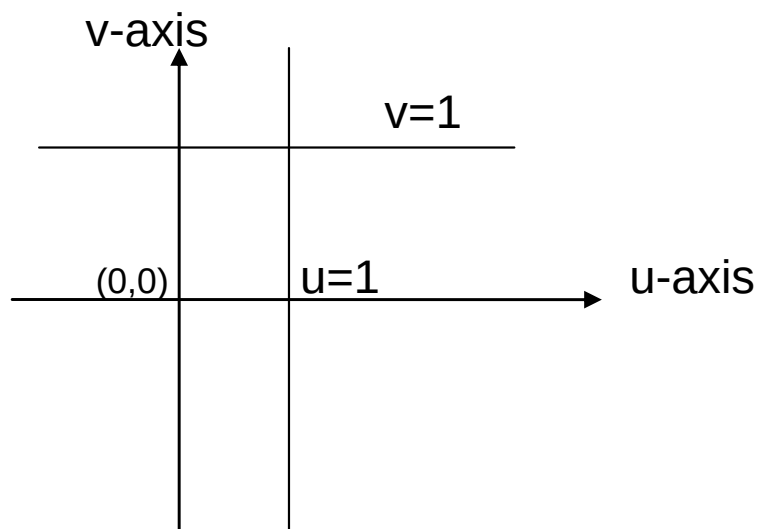
$$x=0 \Rightarrow x = u - y \Rightarrow u-uv=0 \Rightarrow u(1-v)=0 \Rightarrow u=0 \text{ or } v=1$$

$$x=1 \Rightarrow u(1-v)=1$$

X-Y Plane



u-v Plane



Limit of u and v :

$$0 \leq u \leq 1; 0 \leq v \leq 1$$

$$J = \frac{\partial(x, y)}{\partial(u, v)} = \begin{vmatrix} \partial x / \partial u & \partial x / \partial v \\ \partial y / \partial u & \partial y / \partial v \end{vmatrix} = \begin{vmatrix} (1-v) & -u \\ v & u \end{vmatrix} = u$$

Therefore

$$I = \int_{x=0}^1 \int_{y=0}^{1-x} e^{y/(x+y)} dy \cdot dx = \int_{u=0}^1 \int_{v=0}^1 u \cdot e^v du \cdot dv = \frac{1}{2} (e-1)$$

Practice Problems

1. Evaluate by changing to polar coordinates:

a.
$$\int_0^a \int_0^{\sqrt{a^2-y^2}} (x^2 + y^2) dy dx$$

b.
$$\int_0^a \int_y^a \frac{x^2}{\sqrt{x^2 + y^2}} dx dy$$

Ans. (a) $\pi a^4 / 8$, (b) $a^3 \ln(\sqrt{2} + 1) / 3$

2. Evaluate
$$\iint \cos\left(\frac{x-y}{x+y}\right) dx dy$$

Using transformation $u=x-y$; $v=x+y$ over the region bounded by the lines $x=0$; $y=0$; $x+y=1$

Ans: $(\sin 1)/2$

THANK YOU 😊